NIHAL SIDDIQUI

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SUMMARY

AI/ML Engineer specializing in **Retrieval-Augmented Generation (RAG), LLM applications, and production AI systems**. Experienced in building FastAPI services, LangGraph multi-agent workflows, vector databases, and MLOps pipelines using Docker and AWS. Passionate about designing reliable, production-ready AI applications backed by systematic evaluation and measurable improvements.

TECHNICAL SKILLS

Core: Python, SQL, FastAPI, REST APIs, Docker, Git
Generative AI & NLP: LLMs, RAG, LangChain, LangGraph, Pinecone, LangSmith
Machine Learning & MLOps: XGBoost, Scikit-learn, TensorFlow, MLflow, Great Expectations, Optuna, Redis
Cloud & DevOps: AWS (EC2, ECR), GitHub Actions (CI/CD)

PROJECTS

Enterprise-Grade RAG System |GitHub| **Python, FastAPI, LangChain, Pinecone, AWS**

- Built a production-ready RAG API using FastAPI and Pinecone.
- Designed a **hierarchical parent-child chunking pipeline**, improving retrieval relevance from **0.70** → **0.85** through iterative retrieval optimization validated using **LangSmith** and **OpenEvals**.
- Built an automated evaluation pipeline to compare retrieval strategies across **retrieval relevance, correctness, helpfulness, and groundedness**, enabling data-driven optimization of the RAG system.
- Integrated prompt validation, structured logging, LangSmith tracing, and production-ready API architecture for reliable deployment.

Multi-Agent Travel Planner | GitHub | Python, FastAPI, LangGraph, OpenAI, PostgreSQL, Docker

- Built a production-ready multi-agent travel planner using LangGraph with PostgreSQL checkpointing for persistent conversational state.
- Developed FastAPI REST APIs integrated with Tavily Search and custom airport-resolution tools to generate grounded travel itineraries using live information.
- Containerized the application with Docker, integrated LangSmith tracing, structured logging, and Alembic database migrations for production observability and maintainability.

Customer Churn Prediction Platform | GitHub | **XGBoost, MLflow, FastAPI, Docker, AWS EC2**

- Developed an end-to-end churn prediction pipeline achieving **93% recall** through feature engineering and Optuna-based hyperparameter optimization.
- Automated experiment tracking with MLflow and data validation using Great Expectations to ensure reproducible ML workflows.
- Containerized the inference service using Docker and deployed it on AWS EC2 through GitHub Actions CI/CD.
- Optimized classification thresholds using business-driven evaluation to improve recall while reducing false negatives.

EDUCATION

MGM University, Institute of Information and Communication Technology
Bachelor of Technology in Data Science | Expected Graduation: 2027